

A Prospective Falsification-First Architecture for Pure Genus-Two Homology-Cover Renormalization

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Abstract

P32 studies a fixed renormalization proposal for the pure genus-two homology-cover tower $H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. Physical lifted time is scaled by exactly $1/N$, and the logarithm of the raw lift-component product by exactly $1/N^3$. The universal endpoint asks for recovery of every oriented primitive base-owner factor. Adversarial review found that the earlier four-package presentation understated the adverse information already visible in its own prospective local factors. The present article therefore adopts a falsification-first architecture. Conditional on a future derivation of those factors, positive content $d > 1$ and zero content become the first theorem targets; they must be settled before formal or compact-uniform resources are invested in the exceptional content-one stratum. The content-one analytic program is now explicitly contingent and secondary. The project-defined inverse-limit object R_+ , its localizations and projections, and the separate zero-content Hahn object remain unresolved constructions. Likewise, any analytic tail theorem is conditional on survival of the local tests and would require an absolute logarithmic-summand majorant uniform on each frozen compact set and compatible with every specified limit order. The executed work remains literature synthesis over 26 frozen sources and review-informed argument design. No factor, mismatch, obstruction, recovery statement, panel, or limit has been proved or computed. All 26 inherited citations retain `anchor:none`; P32-S13 is bibliographically VERIFIED but remains background-only, so claim-to-passage faithfulness is INCONCLUSIVE. Arithmetic A0 remains unavailable, the formal Route-A tuple is unassigned, positive arithmetic A2 is absent, and Route B remains closed.

繁體中文摘要

本文研究純種屬二同調覆蓋塔的固定重整化方案。基底是帶標記的封閉雙曲曲面及單位速度測地流，所有者涵蓋全部同調內容的保持方向本原共軛類，逆類仍分開處理。物理提升時間嚴格乘以 $1/N$ ，原始提升分量乘積的對數嚴格乘以 $1/N^3$ ；不允許依內容或結果調整。本文採取先否認後一致的架構：只有在未來自足推導局部因子後，才先檢驗正的高內容與零內容；兩者均存活後，才可投入例外的內容一子乘積、形式拓撲與聚集上一致尾項分析。因此內容一只是附條件的次要端點，絕不能代替全所有者回復，也不能移植至另一個殘餘逆極限系統。正內容的逆極限物件、局部化與投影，以及零內容的獨立 Hahn 型物件均尚未構造。本階段僅執行封閉語料綜合與審查後方法設計；未執行所有者綁定、覆蓋計算、因子推導、係數比較、面板、尾界或極限。所有引文均無段落定位，P32-S13 的書目身分已核為 VERIFIED，但仍僅作背景來源，故逐主張支持仍屬未定；本文不宜稱阻礙、回復、新穎性或路線晉級。

Keywords: homology covers; ownerwise factors; falsification-first methods; primitive geodesics; formal products; compact-uniform convergence

中文關鍵詞：同調覆蓋；所有者局部因子；先否證方法；本原測地線；形式乘積；緊集上一致收斂

1 Introduction, Research Question, and Contribution

Finite-cover products can agree on a selected stratum or finite panel without recovering every primitive owner. P32 fixes this distinction at the object level. The base is a marked closed hyperbolic genus-two surface with unit-speed geodesic flow and presentation

$$\Gamma = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle.$$

The cover tower is the pure homology tower $H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. It is not the residual inverse-limit object studied in the preceding branch, and a conclusion about one tower cannot repair the other. Owners are oriented primitive surface-group conjugacy classes; inverse classes remain separately oriented. For an owner with homology vector in $\mathbb{Z}^4$, content d is the greatest common divisor of the absolute coordinates, with zero content treated as a separate case.

The normalization is immutable. Physical lifted time is multiplied by exactly $1/N$, and the logarithm of the raw lift-component product is multiplied by exactly $1/N^3$. Content-dependent changes, outcome-dependent weights, adaptive schedules, and favorable-panel selection are prohibited. The universal endpoint concerns all owner contents, not an average and not the content-one subproduct alone.

The revised research question is:

Do the prospectively derived local factors for higher-content and zero-content owners already obstruct the frozen $1/N$ and $1/N^3$ full-owner recovery scheme; and only if those adverse tests are survived, what formal and analytic obligations remain for the content-one subproduct?

The change is methodological. It does not assert that the candidate factors are correct or that an obstruction exists. Higher- and zero-content checks are earlier in the declared dependency graph because they require fewer named downstream objects than formal topology and compact-uniform tails. This is not an empirical runtime, proof-complexity, or information-gain estimate. A closed adverse result would settle the universal endpoint negatively under the fixed scheme; survival would license only the next predeclared obligation. A positive content-one statement, if eventually proved, would remain restricted.

The closest-work comparison is bounded to the frozen corpus. P32-S01–P32-S06 cover owner algorithms; P32-S07–P32-S12 cover homology covers and lifts; P32-S13–P32-S22 cover products, counting, and tails; and P32-S23–P32-S26 provide formal-algebra background [1–26]. None is used as evidence that the combined dependency order is globally novel, and every passage transfer remains INCONCLUSIVE. The bounded design contribution is the explicit order local-factor derivation, conditional scalar gate, formal-object construction, and compact-uniform tail obligation. No priority or exhaustive originality claim is made.

The word “owner” is universal in this article. It never means only a content-one class. The population comprises oriented primitive surface-group conjugacy classes across positive higher content, content one, and the separately typed zero-content case. Inverse classes remain distinct. This corrects a potentially misleading shorthand in an older state description and prevents a favorable exceptional stratum from being substituted for the frozen all-owner endpoint.

Three gates organize the program. The owner gate must bind equality, oriented conjugacy, maximal roots, inverse handling, serialization, and prefix completeness to the marked presentation. The local-factor gate must derive cover order, component multiplicity, primitive-lift status, rescaled period, and normalized exponent for each relevant content. The formal/analytic gate must supply the coefficient objects and, if local tests survive,

the uniform tail theorem. A failure at an earlier gate stops later work; it is not repaired by changing a schedule, selecting a favorable panel, or moving to a different tower.

The falsification-first order is therefore informational rather than promotional. It asks the shortest exact questions capable of deciding the universal proposal while leaving survival, obstruction, and not-evaluable outcomes all open. The architecture does not infer a mismatch from the displayed shapes of unproved formulas, and it does not infer convergence from a finite diagnostic plan.

2 Frozen Literature and Theoretical Boundary

2.1 Owners, conjugacy, roots, and prefix completeness

The algorithmic corpus supplies several component surfaces. Finite combinatorial structure, word-hyperbolic conjugacy, finite-subset and finite-list decisions, and root-extraction solvability provide adjacent normal-form and decision backgrounds [2–4, 8, 14]. A recent surface-group preprint presents a possible normal-form, conjugacy, and root-finding direction, but it is not peer reviewed and its presentation has not been mapped to the frozen marked presentation [26].

These sources do not bind SG2OwnerCanonical-v1. A complete interface still requires exact equality normal form, oriented conjugacy, maximal primitive roots, separate inverse handling, deterministic serialization, and proof that a requested prefix contains every earlier eligible owner with zero unresolved decisions. Solving a supplied finite list is not the same as certifying prefix completeness.

The distinction between list decision and exhaustion is decisive. A finite-list routine starts from supplied elements; the project prefix must additionally prove that no earlier eligible owner was omitted under the frozen word-length and lexicographic order. Root extraction must be maximal, and orientation must remain external to inversion. Until these properties are proved and replayable, a panel cannot be labeled complete and a zero unresolved count cannot be inferred from a successful run on selected inputs.

2.2 Cover and homology structure

Finite abelian surface covers and Riemannian covering theory supply structural context, while homology-refined geodesic distributions and modern splitting language provide adjacent decomposition surfaces [6, 10–12, 19, 25].

This group can inform deck actions, splitting, and homology strata, but it does not prove P32’s project-specific owner order, primitive lift count, lifted period, or normalized factor. It also does not select $1/N$ or $1/N^3$. Those formulas and choices remain internal theorem targets.

In particular, structural cover vocabulary does not license the project’s displayed local factors. The later derivation must start from the exact deck action of the pure tower, determine the order of the owner image, count and type lifted components, and prove which lifts are primitive. It must then propagate both frozen normalizations without silently moving between raw products and logarithms. These are project-specific proof obligations even when neighboring cover results suggest useful tools.

2.3 Primitive-orbit products and continuation

The historical Selberg record supplies trace and zeta context only. Its bibliographic identity and metadata are VERIFIED, but this article keeps it background-only [22]. Primitive-orbit dynamical-zeta theory, prime-orbit asymptotics, and finite-cover Artin formalism supply general product and factorization contexts [1, 18, 21].

Continuation theory is represented by two routes. The first source is admitted only for the meromorphic-continuation portion left unaffected by its recorded correction; affected spectral-gap and dependent counting claims are excluded. A microlocal route supplies independent continuation context [7, 9].

These works operate at product, asymptotic, or continuation levels. They do not establish independent-owner coefficient equality under the frozen normalization, identify the project formal object with a scalar product, or justify the infinite homology-cover schedule.

2.4 Counting and tail candidates

Renewal methods, prime-geodesic growth, homology-refined orbit counting, and an exponential error-term surface provide possible tail ingredients [13, 15, 20, 23].

The required P32 theorem is stronger than the existence of a nearby asymptotic. It would need an absolute bound for the exact logarithmic summand, uniform on every frozen compact rectangle, compatible with owner strata, and sufficient for every stated iterated and diagonal limit. The frozen verification records do not establish that transfer.

2.5 Formal-topology background

Ordered division rings and well-ordered supports, an inverse limit of countably-variable polynomial rings, an unrelated use of adic completion, and an institutional power-series reference provide four formal or scalar precedents [5, 16, 17, 24].

All four sources remain background-only. Their resemblance to proposed constructions cannot define $\mathbb{R}_+$, its exponent monoid, transition maps, topology, localizations, or singleton projections. They also cannot define the separate zero-content Hahn object or validate the normalization. The frozen comparison CP-P32-004 remains unresolved.

This formal-source boundary is also a truth-condition boundary. A coefficient comparison has no fixed meaning until the coefficient ring, support order, transition maps, localization, and projection are all defined. Resemblance to a known completion or Hahn construction can motivate a definition but cannot supply it. Consequently, the article does not report a coefficient mismatch; it reports the construction and comparison tasks that would make such a statement meaningful.

3 Executed Methodology

3.1 Closed-corpus synthesis

The executed method is a staged synthesis over frozen literature and project records. Phase 2 captured 51 records, removed 12 duplicate manifestations, screened 39 unique records, excluded 13, and retained 26 sources. At that historical Phase-2 checkpoint the identity layer classified 25 sources as VERIFIED and P32-S13 as PLAUSIBLE. A separately authorized Stage-2.5 metadata repair subsequently verified P32-S13's bibliographic identity, so the current identity state is 26 of 26 VERIFIED. The transition does not upgrade its background-only fitness, `anchor:none` status, or INCONCLUSIVE claim-to-passage support. The P32-S02 page correction, P32-S06 preprint boundary, and P32-S17 correction-limited support surface remain unchanged.

The evidence matrix retained all 26 IDs and assigned admitted and excluded claim surfaces. The synthesis grouped them into owner, cover, product, tail, and formal themes. It did not average away publication status, correction status, or background-only roles. It also preserved the distinction between source identity and theorem applicability.

The evidence synthesis was not a vote across papers. Each source-effect row recorded an admitted contribution, an excluded stronger use, and an applicability warning. Thus an algorithmic result could inform one component without being treated as the bound owner interface, and a cover theorem could provide

structural context without proving the project factor. The same discipline keeps P32-S13 background-only and keeps the correction-affected claims of P32-S17 outside the admitted surface.

Citation closure was checked structurally: all source identifiers used in the manuscript resolve to the frozen 26-row inventory and to exactly one bibliography entry. This does not verify any theorem passage. Every use retains `anchor:none`, no direct quotation is introduced, and claim-to-passage status remains `INCONCLUSIVE`. The article therefore uses the literature to bound a research program, not to claim that its project-specific factors have already been derived.

3.2 Revision procedure and nonexecution boundary

The executed scholarly method was a closed-corpus synthesis: records were captured, deduplicated, screened, and grouped under the frozen inclusion frame; source-effect rows separated admitted contributions from excluded stronger uses and applicability warnings; and unresolved source and theorem boundaries were propagated into the research design. Four procedurally separated, single-model-family roles then reviewed the report. Their `MAJOR_REVISION` outcome and the author's adjudication are development provenance, not evidence for a factor, coefficient comparison, obstruction, or limit. Revision 1 changes only the prospective dependency contract and its stated evidence boundary.

The Phase-6 ClaimIntent manifest was frozen before article prose. No new retrieval, source-finalization call, direct quotation, locator, experiment, factor computation, coefficient comparison, panel, or limit was permitted. The executed analytical operation was logical comparison of frozen definitions and review findings. Every scientific disposition remains unexecuted or not evaluated.

4 Review-Adjudicated Proof and Method Architecture

4.1 Frozen candidate factors are theorem targets

For a positive-content owner g , let its content be $d > 0$ and define prospectively $q_N(g) = \gcd(N, d)$. The frozen program asks a later proof to derive the deck order, primitive-lift count, physical lifted period, and normalized local factor, including the target expression

$$F_{N,g}(s) = (1 - \exp(-s \text{ ell}(g)/q_N(g)))^{(-q_N(g))}.$$

For zero content, the frozen program treats the owner separately and targets

$$F_{N,g}^{\wedge(0)}(s) = (1 - \exp(-s \text{ ell}(g)/N))^{(-N)}.$$

Neither expression is a result of this article. A proof must derive them from the deck action under the exact marked presentation, establish the component count and covering degree, prove primitiveness of lifted components, and account for both $1/N$ time scaling and $1/N^3 \log$ multiplicity. The zero-content expression cannot be obtained by informal substitution into the positive-content case.

The base primitive-owner factor used for the proposed comparison is

$$B_g(s) = (1 - \exp(-s \text{ ell}(g)))^{(-1)}.$$

A minimal conditional scalar-check contract is now explicit but unexecuted. Only after an independent derivation of the displayed factor may a future checker fix $\ell > 0$, real $s > 0$, and integer $m \geq 2$, then compare $\Phi_m(s) = (1 - \exp(-s\ell/m))^{-m}$ with $B(s) = (1 - \exp(-s\ell))^{-1}$ under ordinary equality of positive real-valued functions. The proposed substitutions are $m = d$ after higher-content stabilization and $m = N$ for zero content. The comparison has status `UNPROVED/NOT_EVALUABLE` because neither candidate factor has

been independently derived under the frozen $1/N$ time and $1/N^3$ logarithmic normalizations. No inequality, factor mismatch, obstruction, recovery result, or Route credit is asserted here.

The positive-content derivation contract now exposes its intermediate targets. For homology content $d > 0$ and $q_N(g) = \gcd(N, d)$, the proposed identities are deck-image order $N/q_N(g)$, lifted-component count $N^3 q_N(g)$, scaled period $e_{ll}(g)/q_N(g)$, and logarithmically normalized exponent $q_N(g)$. Chaining only those proposed fields gives the displayed F_N, g ; every identity has status UNPROVED. A future ledger must bind the homology vector, content, deck action, component count, covering degree, primitive-lift proof, unscaled and scaled periods, raw exponent, both frozen normalizations, and resulting factor. The zero-content ledger remains separate, and a read-only verifier must reject any unresolved field.

The two normalizations must be visible at different levels. Multiplying physical lifted time by $1/N$ changes the exponential argument, whereas multiplying the logarithm of the raw component product by $1/N^3$ changes its exponent after component counting. Combining those operations informally can hide a missing power or an unjustified primitive-component assumption. The future proof must therefore derive the raw lift product first and apply each frozen normalization in its declared order.

4.2 First adverse target: higher content

The controlling modulus schedule is $N_k = k!$ for every integer $k \geq 1$. For a fixed positive content $d > 1$, the candidate formula would eventually have $q_N(g) = d$ whenever the factorial modulus is divisible by d . The proposed local factor would then have the stabilized displayed shape

$$(1 - \exp(-s \cdot e_{ll}(g)/d))^{(-d)}.$$

The present analysis does not declare this unequal to the base factor in the unresolved project formalism. Instead, it identifies the shortest future theorem: derive the factor self-containedly, construct the permitted singleton-owner comparison, and determine whether equality with B_g holds under the exact frozen notion. One certified mismatch would obstruct universal full-owner recovery for the fixed scheme. A failed derivation would stop the program earlier as not evaluable.

This priority is falsification-first because it can decide the universal endpoint before a global infinite product is built. It also prevents a content-one success from concealing an adverse owner class.

The higher-content test is ownerwise and target blind. A later execution would have to select an eligible owner under the frozen owner interface, not because its displayed length or content appears favorable. It would then derive the factor and project to the independently named owner variable in the completed positive-content formal object. Only a replayable inequality under that object's exact equality relation could be labeled an obstruction. An unresolved projection or factor derivation returns a not-evaluable state instead.

Survival also has a narrow meaning. If the first registered higher-content comparison does not yield a mismatch, that result does not establish the universal scheme, the formula for every owner, or an infinite product. It licenses only the next predeclared test or proof obligation. The architecture thereby avoids turning a failed attempt at falsification into a positive recovery theorem.

4.3 Second adverse target: zero content

Zero content has order one in the prospective cover algebra, N^4 candidate lifted components, rescaled period $e_{ll}(g)/N$, and normalized exponent N . Each of those fields requires proof. Conditional on that derivation, the zero-content factor is the second local comparison target. It must be represented in the separately typed zero-content formal object and compared with the base owner factor without pretending that $d=0$ belongs to the positive-content exponent system.

No fixed- N coefficient mismatch is computed here. The article records only the obligation: derive the factor, specify the zero-content formal semantics, perform the first permitted coefficient comparison, and return

either survival, a certified ownerwise obstruction, or a typed not-evaluable status. The review's face-value concern is a reason for priority, not evidence that the scientific endpoint has already failed.

Zero content requires this separate path because its homology image has a different algebraic status from a positive integer content. Treating $d = 0$ as a limiting or substituted positive value would obscure the component and exponent arguments. The zero-content certificate must state its own support monoid, factor semantics, and comparison map. It must also preserve the same oriented primitive-owner policy used by the other strata.

The accepted outcomes are deliberately three-valued. A closed derivation followed by equality records local survival; a closed derivation followed by certified inequality records an ownerwise adverse result for the frozen scheme; and an unclosed derivation or formal comparison records not evaluable. None of these outcomes has been produced here. In particular, the face-value appearance of the prospective exponent N is not a computed coefficient certificate.

4.4 Content one is contingent and secondary

For $d=1$, the candidate expression has $q_N=1$, making this stratum structurally exceptional within the proposed formula. That observation does not establish recovery. A content-one program still requires canonical owner enumeration, a well-defined formal product, scalar specialization, absolute compact-uniform control, cofinality, and justified limit interchange.

The branch may begin only if the higher- and zero-content theorem targets are resolved in a way that warrants further work. Even a future positive content-one result would describe a restricted subproduct. It would not reinstate universal owner recovery, repair the residual inverse-limit system, supply arithmetic A0, earn automatic positive A2 credit, or open Route B.

Content one remains scientifically interesting because the candidate formula simplifies there, but simplification is not recovery. The branch still needs a prefix-complete enumeration over oriented primitive owners, a product defined in the chosen formal topology, a justified scalar specialization, and a compact-uniform logarithmic tail. A positive statement about that subproduct must name the restriction in its title, theorem, and data surfaces. It cannot be relabeled as an all-content primitive product.

Nor can the pure tower repair the earlier residual inverse-limit object. The systems have different owners and quantified limits. Even a complete content-one theorem for the pure tower would remain a calibrator result under the fixed $1/N$ and $1/N^3$ scheme. This separation is part of the initial dynamical restriction, not a caveat to be reconsidered after outcomes.

4.5 Formal objects remain unresolved

The positive-content comparison proposes independent owner variables, a directed inverse-limit object R_+ , localizations, transition maps, and finite-support singleton projections. The zero-content comparison proposes a separate nonnegative-rational Hahn-type object. The present article does not define these objects by importing a literature analogy. A valid later construction must state the exponent monoids, coefficient rings, support conditions, directed maps, topology, localization domains, and projection compatibility, then prove every operation used by the factor comparison is well defined.

Object or map	Intended typed surface	Status and prerequisite
R_+	Positive-content factors; coefficient ring, exponent monoid, support and topology	UNDEFINED; closed factor derivations
transition/localization	Directed maps and declared denominator domains	UNDEFINED; compatible R_+
R_0	Separate zero-content Hahn-type support and equality	UNDEFINED; closed zero-content factor
π_g	Singleton owner projection with stated domain and codomain	UNDEFINED; transition compatibility
scalar specialization	Formal object to scalar function on a declared domain	UNDEFINED; local-survival receipt
majorant	Exact logarithmic summands to a summable compact-uniform bound	UNPROVED; surviving scalar branch
limit claims	Iterated and diagonal registry rows	UNPROVED; majorant and interchange

No row records a constructed object, completed operation, or scientific result.

CP-P32-004 remains `flagged_unresolved`. The coefficient rings, exponent monoids, support conditions, transition maps, topologies, localization domains, equality relations, and singleton projections for R_+ have status UNDEFINED; the corresponding fields for the separate zero-content Hahn-type object have the same status. Consequently coefficient and singleton-projection comparisons are NOT_EVALUABLE. No formal object or operation is imported by analogy, and no coefficient mismatch is used as a completed scientific finding.

For the positive-content object, a later definition must make finite-support coefficient projections compatible with every transition map. If localization is used, the denominators and their domains must be specified before factors are compared. For the zero-content Hahn-type object, the ordered exponent group and well-ordered support condition must be stated independently. A map between the two objects cannot be assumed merely because both contain symbols resembling exponentials.

Singleton projection is central to the falsification-first strategy: it is intended to isolate one owner without requiring construction of the whole scalar product. But the projection itself needs a theorem. It must be defined on the exact completed object, commute with the relevant transition maps, and preserve the equality notion used in the adverse comparison. Until those properties close, a visual difference between two formal expressions has no registered evidentiary force.

4.6 Analytic tail work is survival-conditional

Only a surviving content-one formal branch reaches the analytic stage. The scalar theorem target would need to state exact functions, summation order, owner exhaustion, modulus schedule, compact domain, and limit order. The frozen compact rectangles are

$$K(\delta, T, R) = \{s : 1 + \delta \leq \operatorname{Re}(s) \leq R, \quad |\operatorname{Im}(s)| \leq T\},$$

for every $\delta > 0$, finite $T \geq 0$, and finite $R \geq 1 + \delta$. The schedules remain

$$N_k = k!, \quad m_k = 2^k, \quad N'_k = 2 \cdot (k!), \quad k \geq 1.$$

Write $L(m, N; s)$ only as a registry name for the prospective ordered sum of the first m exact logarithmic owner summands; it is not yet instantiated because the factors are unproved. For each compact $K(\delta, T, R)$ and each schedule S_k in $\{N_k = k!, N'_k = 2 \cdot (k!)\}$, the wholly UNPROVED registry contains: AN-1, owner exhaustion $m \rightarrow \infty$ at fixed S_k ; AN-2, schedule growth $k \rightarrow \infty$ at fixed m ; AN-3, the iterated order $k \rightarrow \infty$ after $m \rightarrow \infty$; AN-4, the reverse iterated order; and AN-5, the diagonal

$L(m_k, S_k; s)$ with $m_k = 2^k$. Every row requires the exact summand definition, an absolute summable majorant uniform on the compact and independent of the permitted indices, the coupling rule, and a named interchange statement. The prefix $1 \leq k \leq 8$ and panel sizes m in $\{8, 16, 32, 64, 128\}$ remain unexecuted diagnostics with no convergence force.

The order of limits is part of the claim. A theorem for fixed panel size followed by modulus growth does not automatically imply the reverse order or the registered diagonal. Likewise, uniformity on sampled points does not imply uniformity on every compact rectangle. The future majorant must dominate the exact logarithmic summands independently of the allowed owner prefix and modulus indices in the stated range, and its summability must justify every interchange actually claimed.

The factorial and neighboring schedules act as fixed stress paths. They may not be shortened after an inconvenient content appears, and the finite prefix through $k = 8$ remains only a prospective replay and serialization diagnostic. No observation on that prefix, even exact, would establish cofinality, convergence, analytic continuation, or a limit identity. This keeps numerical or finite symbolic checks in their proper evidence class.

5 Evidence-Synthesis Findings

Eight frozen positions govern the article. First, the algorithmic corpus supplies separate normal-form, conjugacy, root, list-decision, and serialization candidates but does not bind `SG2OwnerCanonical-v1`. Second, higher-content and zero-content formulas are adverse local theorem targets conditional on a future self-contained derivation. Third, the program is falsification-first: those targets precede formal or compact-uniform investment in content one.

Fourth, a content-one subproduct is contingent and secondary. Fifth, R_+ , its localizations and projections, and the separate zero-content object remain construction obligations because CP-P32-004 is unresolved. Sixth, only after local survival would the analytic branch require compact-uniform absolute tail control compatible with every frozen limit order. Seventh, the exact schedules remain unchanged and finite prefixes remain unexecuted diagnostics. Eighth, the revision concerns only the pure tower and changes no scientific recovery or Route status.

These are literature-synthesis and research-design findings. They do not constitute factor derivations, coefficient certificates, obstruction theorems, convergence results, or novelty findings.

6 Reproducibility and Prospective Interface

The present phase is reproducible at the artifact level. The source inventory, verification rows, Phase-4 report, four Phase-5 reviews, synthesis, checkpoint, frozen research contract, and ClaimIntent manifest are hash-bound. The 26 citation IDs close against the 26 references and verification rows. This does not make the prose generation deterministic or the theorem transfers passage-verified.

A later scientific interface should freeze producer and verifier roles before outcomes. The owner layer must bind the exact marked presentation, equality, oriented conjugacy, maximal roots, inverse policy, serialization, and prefix completeness. The cover layer must bind deck action, component count, primitive-lift proof, physical period, and normalization. The formal layer must bind the positive and zero-content objects separately. Only after those bindings pass should the adverse singleton comparisons run.

The smallest next package is therefore not an owner panel or numerical grid. It is a proof-binding dossier for one higher-content owner and one zero-content owner selected without target inspection. The dossier should expose every algebraic premise and return a typed result: derivation closed and comparison eligible, derivation failed and not evaluable, or later certified mismatch/survival after separately authorized scientific execution.

That dossier should be reproducible without access to hidden producer state. It should include the frozen input and presentation hashes, the owner certificate, homology vector and content, theorem and schema versions, deck-action derivation, primitive-lift evidence, normalization trace, formal-object version, projection proof, verifier receipt, and a typed disposition. Positive and negative cases should use the same field contract. Missing evidence should invalidate the comparison rather than be represented by a default value.

Only after the two local dossiers pass their interface checks would a population manifest be meaningful. Such a manifest would freeze the target-blind enumeration rule and exact schedules before any factors are read. This separation prevents implementation debugging, owner selection, and scientific interpretation from occurring in one outcome-dependent loop. None of these dossiers or manifests exists in the present manuscript.

7 Discussion and Implications

The falsification-first order changes the value proposition of P32. The earlier design placed owner binding, lift algebra, formal construction, and analytic tails in one broad progression. The revised architecture recognizes that not every package is needed to test the universal claim. If higher or zero content fails under an exact local-factor theorem, the full-owner endpoint is settled for the frozen normalization before the content-one tail problem becomes relevant.

This does not prejudge the outcome. The candidate factors are themselves unproved, and the formal truth conditions are unfinished. The method therefore has two early stopping surfaces: failure to derive the factor, or a later certified adverse comparison. Both are scientifically clearer than continuing to an infinite product while the local ingredients remain unresolved.

Content one still has potential value as a restricted analytic object. Its exceptional status must remain visible. A favorable theorem there would not erase an ownerwise obstruction elsewhere, and it would not create arithmetic specificity in a generic homology-cover calibrator. The pure tower has no intrinsic rational-prime owner map or $\log p$ clock in this record.

The Route boundary follows directly. A0 is unavailable, so P32 is not a primary Hilbert–Pólya dynamics candidate. No stable dynamical determinant or divisor comparison has been produced, A2–A4 remain unrun, and no Route-B entry condition exists. A negative local theorem could still be a meaningful calibration result, but it would not be Route promotion.

8 Acknowledged Limitations

All 26 prose citations retain `anchor:none`. The frozen workflow records aggregate capture and screening counts and the retained-source effects, but it does not contain a complete 51-record inclusion/exclusion decision log or exact theorem-page, hypothesis, and quotation locators for every transfer. Claim-to-passage faithfulness therefore remains INCONCLUSIVE. No retrieval or source-finalization was authorized for this revision, so missing queries, decisions, and passages are reported as unavailable rather than reconstructed after the fact.

P32-S13 is bibliographically VERIFIED but remains background-only. P32-S06 remains a non-peer-reviewed preprint whose presentation mapping is unproved. P32-S17 remains restricted to the correction-unaffected continuation surface. The formal cluster remains background-only. General retraction and source-conflict/COI checks remain NOT_CHECKED and UNKNOWN_NOT_CHECKED.

The core mathematical objects are unresolved. `SG2ownerCanonical-v1` is unbound. The positive- and zero-content factors are derivation targets. R_+ , its maps and projections, and the separate zero-content object are not closed constructions. No compact-uniform majorant or limit theorem exists in the project artifacts. No diagnostic was run.

The corpus and article also do not establish novelty. The output is deliberately positioned as a bounded research-program study. AI-assisted roles used one Codex model family with procedural separation rather than statistically independent validation. Human gate approval does not substitute for source-passage reading or mathematical proof.

9 Future Work

Future work should preserve the falsification-first order:

1. perform a separately authorized source-finalization pass for exact theorem passages, hypotheses, correction state, and any required retraction/source-conflict checks;
2. bind `SG2OwnerCanonical-v1` to the exact marked presentation with an independent verifier and prefix-completeness theorem;
3. derive the higher-content and zero-content owner factors from the deck action, including primitive-lift status and both fixed normalizations;
4. construct the positive-content and zero-content formal objects separately and prove singleton projection validity;
5. execute one target-blind higher-content and one zero-content comparison only under new scientific authorization;
6. stop the universal program upon a certified adverse result, or document survival without promoting it to recovery; and
7. only after survival, formulate and prove the content-one compact-uniform tail theorem on every frozen schedule and limit order.

The finite diagnostic prefix may later test serialization and replay. It cannot establish a limit and should not be scheduled before the relevant proof surfaces close.

10 Conclusion

P32 now places its shortest discriminating obligations first. Higher-content and zero-content owner factors are the primary conditional theorem targets. The content-one formal and analytic program is secondary and may proceed only after the adverse local tests survive. The formal objects remain unresolved, and tail work remains conditional on both local survival and a self-contained scalar theorem.

This manuscript is a design-level advance, not a mathematical obstruction or recovery result. No owner interface, factor derivation, coefficient test, panel, compact-uniform bound, limit, or scientific computation was executed. All citation passages remain unresolved, P32-S13 is bibliographically VERIFIED but remains background-only, arithmetic A0 remains unavailable, the formal Route-A tuple remains UNASSIGNED, positive arithmetic A2 remains absent, and Route B remains closed.

Declarations

Author contribution and accountability. Liang Wang specified the inherited mathematical object and its restrictions, supplied the conceptual direction, approved the recorded stage gates, adjudicated the falsification-first design, and is responsible for the manuscript’s scholarly content. AI assistance is disclosed below and is not credited with authorship. Stage-gate approval does not attest that the author personally performed full-text or exact source-passage verification.

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Ethics statement. Human-subjects and animal-research review is not applicable. The work concerns theoretical mathematics, published-source records, and project-owned workflow artifacts; it involves no participants, identifiable personal data, animals, recruitment, or intervention.

Data and materials availability. Current materials are resolvable only by repository-relative project paths and content digests:

- `notes/stage1_phase2_source_inventory.tsv`
(c375c1e7e8310d6d5a1aa4509147a2d4b61b75fe3283b6d148fd0c61f2e76d8e)
- `notes/stage1_phase2_source_verification.tsv`
(7ab194dd2a26a53aa8bc2908c91048fc10b682b73ddf6d93023ef5b94e865e08)
- `notes/stage2_5_material_passport.json`
(f3136c99aab75685063e5fc8ea61dfe814c333fd60797a39690d802af4788cae)
- `notes/stage3_revision_roadmap.json`
(e2fd60e6344abba81714096a3d0c60fd0522da853fa54f83d002536fbfb470c8)

These are internal locators, not a deposited archival release or persistent public identifier. No owner panel, cover computation, coefficient table, tail dataset, factor certificate, or scientific result was generated during manuscript preparation.

AI Disclosure and Verification Limitation

OpenAI Codex, using the GPT-5 model family, assisted during the session dated 2026-09-02 UTC; the exact backend snapshot/build was not exposed. Recorded AI-assisted work included literature-search support, source-identity and metadata checking, evidence-matrix construction, evidence synthesis, report drafting, four role-based Phase-5 reviews, review synthesis, ClaimIntent-constrained Revision-1 drafting, citation-ID/reference closure checks, source-status preservation checks, and stable-ID revision accounting. No AI system executed an owner interface, cover calculation, factor derivation, coefficient test, numerical panel, limit proof, experiment, obstruction certificate, or canonical-results refresh.

Liang Wang is the responsible human author. He approved the project restrictions, workflow gates, and the Phase-6 falsification-first design choice. Those approvals do not state that he personally read every source in full or verified any claim at the exact source-passage level. The recorded source work was bounded mainly to identity, metadata, abstracts, authoritative landing pages, correction records, and project-local claim-fitness judgments. All 26 citations lack passage locators; claim-to-passage faithfulness remains INCONCLUSIVE. The article does not claim theorem-level source verification, novelty, impossibility, or a clean retraction/conflict screen.

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